

REPORT ON THE ADVANCED INDUSTRIAL AUTOMATION WORKSHOP (PLC, SCADA, HMI & VFD)

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1. Introduction

The Advanced Industrial Automation Workshop conducted from **26th to 30th January 2026** by the **Department of Electronics and Instrumentation Engineering, SRM Institute of Science and Technology**, was one of the most useful and genuinely enriching technical experiences of my M.Tech journey so far. The workshop focused on PLC, SCADA, HMI, and VFD, with the clear purpose of giving students practical exposure to industrial automation systems and helping them bridge the gap between classroom learning and real industrial practice.

For me, this workshop was especially relevant because industrial automation is part of my current semester, and it also aligns closely with my long-term interest in Automation and Robotics. SRMIST offers programs in **both Automation and Robotics and Electronics and Instrumentation Engineering**, which makes this kind of workshop highly relevant to the academic environment around me. (SRM Institute) As a student who shifted from a Computer Science background with AI and ML exposure into automation, I have been learning to think differently. Software logic is familiar territory, but industrial systems demand something more: patience, wiring awareness, process understanding, hardware discipline, and practical troubleshooting.

This workshop helped me grow in that direction.

What I appreciated most was that it did not feel like a flashy event meant only for certificates and photographs. It felt real. The sessions were **grounded**, the explanations were **practical**, and the overall learning atmosphere was **open and supportive**. It gave me the kind of exposure that textbooks alone cannot provide.

2. Purpose and Relevance of the Workshop

The workshop brochure clearly mentioned that the program was designed to bridge the gap between academic learning and industrial practice through hands-on exposure to modern automation technologies used in real industrial environments. It also highlighted the importance of PLC programming, SCADA system integration, HMI design, and VFD applications in developing technical competency and industry readiness.

This purpose made a lot of sense to me.

In a classroom, industrial automation often feels like a subject with many terms, diagrams, and processes. But when you actually see how these systems work together, the subject starts making sense as a complete engineering domain. PLCs control logic, SCADA supervises and

monitors, HMI gives the operator an interface, and VFDs manage motor speed and direction. Suddenly, the field stops looking like disconnected theory and starts looking like a functioning system.

That is exactly what this workshop did for me.

It connected theory to practice in a way that was simple, effective, and easy to remember. Since this topic is also part of my semester, the workshop became more than just an extra learning activity. It became an important support for my current academic work.

3. Workshop Environment and Teaching Style

The environment during the workshop was one of its strongest points.

The sessions were handled throughout the week by **Mr. Himanshu Kumar**, Director, INDWELL Automation. His teaching style was one of the main reasons the workshop was so effective. He explained concepts in a practical, calm, and approachable manner. He did not rush through the topics, and he did not speak in a way that made students feel overwhelmed. Instead, he took the time to explain the reasoning behind each process and each control sequence.

That mattered a lot.

A workshop becomes useful only when students are comfortable asking questions. In this case, the atmosphere was open enough that even basic doubts could be asked without hesitation. The faculty members and resource person were friendly, patient, and willing to clarify even repeated questions. That kind of openness makes a huge difference in technical learning because students do not learn well when they are afraid to speak.

What I personally liked was that the workshop felt interactive rather than one-sided. It was not simply a lecture series. It was a space where students could think aloud, ask “why” and “how,” and receive practical answers. The learning became more meaningful because of that.

4. Learning PLC Programming in a Practical Way

Among all the topics covered, the **PLC sessions** were the most valuable for me personally.

Before this workshop, ladder logic felt a little unfamiliar. Coming from a coding background, I naturally tried to compare it with conventional programming languages, but PLC logic works in a very different way. It is not just about code. It is about process control, sequence management, reliability, and machine behavior.

During the workshop, we learned about PLC hardware, software, addressing, timers, counters, memory flags, and ladder logic development. My notes included practical references to Mitsubishi PLC systems, input and output addressing, timers, counters, and communication setup. The workshop brochure also specifically mentioned Mitsubishi PLC, Codesys practice, and project development as part of the PLC module.

What made this part so useful was the way the logic was explained step by step. We were not just shown symbols on a board. We were shown how industrial operations are translated into ladder sequences. The examples included motor control, sequential operations, timer-based actions, repeating cycles, and process control logic.

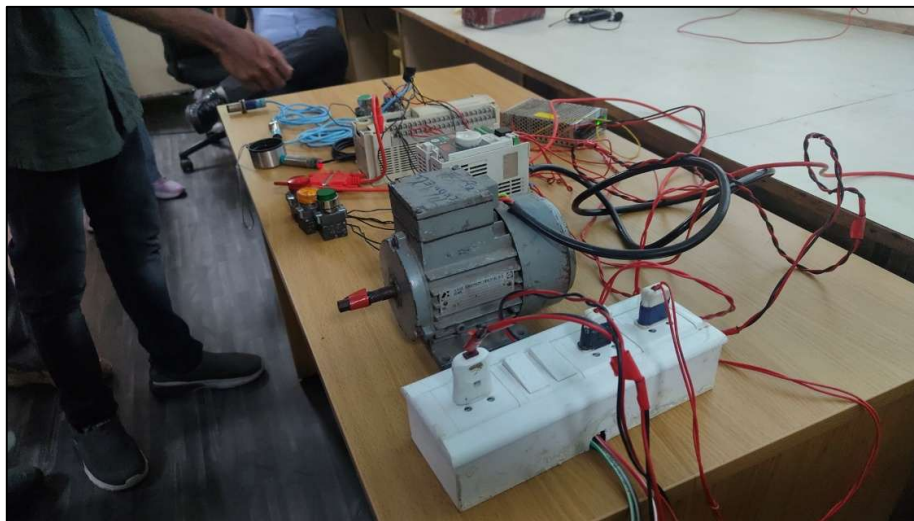
At first, ladder diagrams may look simple. But once you start tracing the sequence carefully, you realize how much thought goes into even a basic industrial control process.

I also understood something important during these sessions: in industrial automation, a small mistake can change the entire behavior of a system. A wrong contact, a missing condition, or an incorrect timer logic can affect the whole sequence. That makes precision essential. It also teaches a certain discipline that is very useful for future automation and robotics work.

5. Hands-On Exposure and Practical Understanding

The practical sessions were one of the best parts of the workshop.

We were exposed to industrial hardware setups involving PLCs, motors, VFDs, wiring systems, switches, and communication interfaces. The brochure mentioned that participants would gain practical insights into PLC programming, SCADA integration, HMI design, and VFD applications through expert-led sessions, and that was exactly the kind of experience the workshop delivered.



Seeing the motor respond to control logic made a huge difference in my understanding. Reading about motor control in theory is one thing. Watching the actual system respond to a VFD setting or PLC command is something else entirely. It made the concept feel alive.

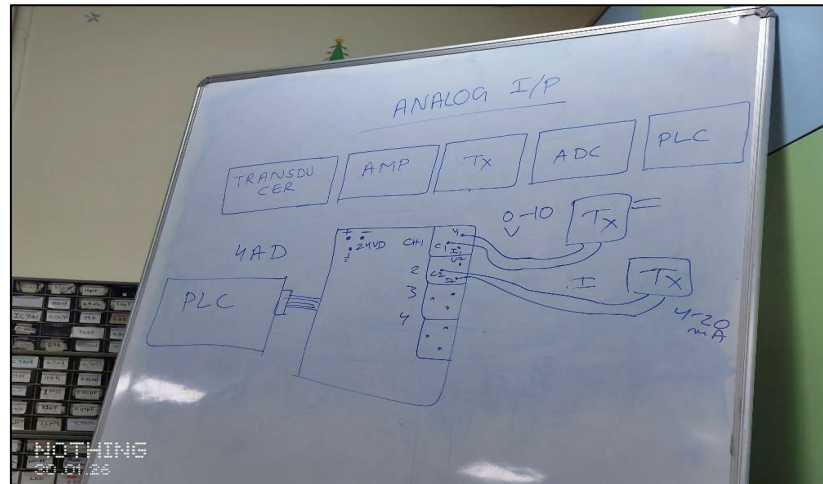
We also learned practical procedures like:

- establishing PLC communication,
- checking COM port settings,
- using GX Developer and GX Works,

- performing online transfer,
- and debugging communication issues.

These may sound like small procedural details, but in reality, they are extremely important. Industrial systems often fail because of basic setup errors, not because the concept itself is difficult. Learning those steps in a practical environment made me more careful and more aware of how real industrial work is done.

The workshop also covered sink and source wiring, SMPS connections, and analog input handling. One of the whiteboard explanations showing the chain:



Transducer → Amplifier → Transmitter → ADC → PLC

was very helpful. It simplified the signal flow and helped me understand how industrial input systems are structured.

6. Learning SCADA, HMI, and Industrial Monitoring

Another useful part of the workshop was the introduction to **SCADA** and **HMI** systems.

Before this workshop, I understood these terms only at a basic level. But the sessions showed that SCADA and HMI are not just add-ons to a control system. They are essential parts of industrial operation.

SCADA allows monitoring, supervision, and data handling. HMI allows human operators to interact with the system through screens, indicators, and controls. The brochure mentioned topics such as SCADA interfacing with PLC, screen development, digital and analog data configuration, analog data scaling, driver selection, tag creation, password levels, and project development. It also mentioned HMI screen development and industrial project work.

This was useful because it showed the human side of automation. Machines may do the work, but people still need to monitor them, check values, detect faults, and make decisions. SCADA and HMI make that possible.

The workshop helped me see how industrial systems are not just about control in the background. They are also about visibility, supervision, and operator awareness. That perspective was important for me because it connected automation with interface design and real-time monitoring, which are both very relevant in modern industrial systems.

7. Learning VFD-Based Motor Control

The **VFD section** was another highlight of the workshop.

The brochure specifically mentioned that the workshop would cover AC drive basics, parameter setting, speed control, direction control, acceleration and deceleration time control, PLC interfacing, and multispeed or multidirectional motor control projects.

This part helped me understand how an induction motor can be controlled in a flexible and efficient manner. Speed control is not just about “making a motor run slower or faster.” It is about controlling motion properly according to the application. That matters a lot in conveyors, pumps, fans, manufacturing systems, and robotics-related actuation.

As a student moving into robotics and automation, this session was especially meaningful. Motion control is a key part of automated systems. Learning how motors respond to drive control gave me a better foundation for future work in robotics, embedded motion systems, and industrial automation projects.

What I appreciated here was that the explanation was not overly mathematical or unnecessarily complicated. It was simple enough to follow, but still detailed enough to be useful. That balance made the learning effective.

8. The Value of the ISA Community

Another important part of the workshop was the involvement of **ISA South India Section** members.

They spoke about the importance of:

- being part of a technical community,
- interacting with professionals,
- continuing to learn beyond the classroom,
- attending workshops and events,
- and staying connected to industry practices.

This was a meaningful message.

As students, it is very easy to think that learning ends with subjects, exams, and grades. But the truth is that engineering growth happens continuously. A community helps in that process. It exposes students to standards, networking, updates, and real-world discussions that classroom teaching may not always cover.



That message stayed with me because it was honest and practical. It reminded me that growth in a technical field is not only about marks. It is also about staying curious and staying connected.

9. Personal Reflection on the Experience

From my own perspective, this workshop was useful in a very direct way.

It helped me understand the subject better because industrial automation is part of my semester. It also helped me feel more confident about a field that I am still learning. That is something I value a lot.

Before attending, some of these concepts felt distant and difficult. After the workshop, they felt more structured and much more understandable. I am still learning, but I no longer feel as though the subject is outside my reach.

The workshop also showed me something important about engineering learning: not everything can be learned from notes or slides. Some things only become clear when you see them in action, ask questions, and slowly work through the practical details. That is exactly what happened here.

I also appreciated the fact that the teaching was not exaggerated or unrealistic. It was grounded. It respected the fact that students are still learning, and it gave us enough space to ask, observe, and understand properly.

That kind of learning environment is rare and very valuable.

10. Conclusion

The **Advanced Industrial Automation Workshop** conducted from **26th to 30th January 2026** was a highly useful, practical, and intellectually enriching experience for me as an M.Tech student. It successfully combined theory with hands-on exposure in **PLC, SCADA, HMI, VFD, motor control, industrial communication, and instrumentation**. The workshop's purpose of bridging academic learning with industrial practice was clearly achieved.

The sessions conducted by **Mr. Himanshu Kumar** were especially valuable because they were practical, approachable, and easy to follow. The open and friendly attitude of the faculty made the workshop comfortable for students and encouraged real interaction. The insights shared by ISA South India Section members added another layer of value by highlighting the importance of professional communities and lifelong learning.

For me personally, the workshop was more than just a technical program. It helped me connect my semester subject with actual industrial systems, strengthened my understanding of automation, and gave me a better idea of how theory becomes practice in the real world.

Most importantly, it made the field feel more approachable.

And that, honestly, is a big deal.



THANK YOU!